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§10. Rings, ideals, quotient rings.

DEF A ring R is a set of together with two binary operations $+$ and $\cdot$, st.

1). $(R,+)$ is an abelian group with 0 the additive unit.

2). $\cdot$ is associative.

3). the distribution law holds.

☆ 4). R is unital, ie. $\exists$ element $1\in R$, st. $1\cdot a=a\cdot 1=a$, $\forall a\in R$.

"rng" without 1.

$1\neq 0$ ↕

DEF A nonzero ring is called a division ring (除环) or skew field if every nonzero element $a\in R$ has a multiplicative inverse (ie. $\exists a^{-1}$, $aa^{-1}=a^{-1}a=1$).

A commutative division ring is called a field.

eg. 1) $\{0\}$, $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$.

2) $\mathbb{Z}[\frac{1}{N}]:=\{\frac{a}{N^r}\in\mathbb{Q}\mid a\in\mathbb{Z}, r\in\mathbb{Z}_{\geq 0}\}$ is a subring of $\mathbb{Q}$.

3) if R is a commutative ring

$R[x]:=\{\sum_{n\geq 0}a_nx^n \mid a_n\in R, a_n=0 \text{ for all but finitely many } n\}$ is a commutative ring.

more generally $R[x_1,\dots,x_m]$ $R[x_j]_{j\in J}$.

4) If R is a ring, then $\mathrm{Mat}_n(R)$ is a ring.

5) $\mathbb{H}=\{a+bi+cj+dk \mid a,b,c,d\in\mathbb{R}\}$ Hamiltonian quaternions

($\mathbb{H}$ ⊃ $\mathbb{C}$ ⊃ $\mathbb{R}$)

multiplication. $i^2=j^2=k^2=ijk=-1$.

conjugation. $\overline{a+bi+cj+dk}=a-bi-cj-dk$

norm. $|z|^2=z\bar{z}=a^2+b^2+c^2+d^2\in\mathbb{R}_{\geq 0}$.

inverse. if $z\neq 0$, $z^{-1}:=\frac{\bar{z}}{|z|^2}$

$\Rightarrow\mathbb{H}$ is a noncommutative division ring.

6) Group ring. Let R be a (commutative) ring and G a group.

Define $R[G] := \{\sum_{g\in G} a_g g \mid a_g \in R,\ a_g = 0 \text{ for all but finitely many } g\}$.

multiplication $\left(\sum_{g\in G} a_g g\right)\left(\sum_{h\in G} b_h h\right) := \sum_{g,h\in G} (a_g b_h)(gh)$

eg. R commutative $G=\mathbb{Z}$ $R[\mathbb{Z}] = R[x,x^{-1}] = \{\sum_{n\in\mathbb{Z}} a_n x^n \mid a_n \in R,\ a_n = 0 \text{ for all but finitely many}\}$.

DEF. Let R and S be rings.

A map $\Phi: R \to S$ is called a homomorphism of rings if

1). $\forall a,b\in R,\ \Phi(a+b) = \Phi(a)+\Phi(b)$ $(\Rightarrow \Phi(0_R) = 0_S)$

2). $\forall a,b\in R,\ \Phi(ab) = \Phi(a)\Phi(b)$ → 行意环对乘法并不一定构成群，故额外地需要3)

✱3). $\Phi(1_R) = 1_S$.

Φ is an isomorphism of rings if moreover Φ is bijective.

eg. · $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$

· if R is commutative
$\forall a\in R$ $\Phi_a: R[x] \to R$
$f(x) \mapsto f(a)$. "代入"

DEF. Let R be a ring.

1). an element $a\in R$ is called a left(right) zero-divisor if

$\exists\, 0\neq b\in R$, s.t. $ab=0$ (respectively $ba=0$)

2). an element $u\in R$ is called a unit in R if

u has a multiplicative inverse.

$R^\times := \{\text{units in } R\}$ multiplicative group.

为何要求零环不是整环？因为我们希望1是素数

3). A nonzero commutative ring R containing NO nonzero zero-divisor is called an integral domain. 整环

$(\Leftrightarrow$ cancellation law. $ab=0 \Rightarrow a=0$ or $b=0$. $\quad a\neq 0, b\neq 0 \Rightarrow ab\neq 0.)$

eg. · $(\mathbb{Z}/n\mathbb{Z})^\times = \{a \bmod n \mid \gcd(a,n)=1\}$

zero-divisors in $\mathbb{Z}/n\mathbb{Z} = \{a \bmod n \mid \gcd(a,n)\neq 1\}$.

· if R is an integral domain, so is $R[x]$. 看首项即可.

LEM A finite integral domain is a field.

Pf: want $\forall 0\neq a\in R$, a has a multiplicative inverse.

Consider $m_a: R\to R$, $x\mapsto ax$ homomorphism of additive group

$\ker(m_a)=\{x\in R \mid ax=0\}=\{0\}$ (整环) $\Rightarrow m_a$ is injective $\overset{R \text{ finite}}{\Longrightarrow}$ m_a is surjective then $\vec{a}=m_a(1)$. $\square$

DEF Let R be an integral domain. (形式化地做成分数)

Define the fraction field of R.

$\mathrm{Frac}(R)=\left\{\frac{a}{b} \mid a\in R, b\in R\setminus\{0\}\right\}\Big/ \frac{a}{b}\sim\frac{c}{d}$ if $ad=bc$

$\mathrm{Frac}(R)$ is a field.

eg. $\mathrm{Frac}(\mathbb{Z})=\mathbb{Q}$ $\mathrm{Frac}(F[x])=F(x)$ (field of rational functions)

Ideals, quotient rings (为了作商而引入, 对比反规子群)

Question. how to define quotient for rings?

Want: identify a subset of R with 0.

"吸收" $a\cdot 0=0\cdot a=0$ $\forall a\in R$ quotient.

DEF A subset $I\subset R$ is called a left(right) ideal if

1). $\forall a,b\in I$, $a-b\in I$ (additive subgroup)

2). $\forall a\in I, r\in R$, $ra\in I$ (respectively $ar\in I$).

$R\curvearrowright I$ left $I\curvearrowleft R$ right

I is called an ideal or two-sided ideal if it is both a left ideal and a right ideal.

in particular, if R commutative, then left ideal = right ideal = ideal.

! An ideal is usually not a subring as we do not necessarily have $1\in I$. ($1\in I\Leftrightarrow I=R$)

eg. trivial ideals: $\{0\}$ and R.

DEF Let R be a ring and $I\subset R$ a two-sided ideal.

Define the quotient ring $R/I=\{r+I \mid r\in R\}$ (additive cosets)

$(r+I)+(s+I):=(r+s)+I$ $\quad (r+I)\cdot(s+I):=(rs)+I$.

Can check $\cdot$ is well-defined. 恰好用到理想的定义，也说明理想的引入是合理的.

There is a natural surjective homomorphism

$\pi: R \to R/I$ "projection". → 0的原像.
$r \mapsto r+I$ with $\ker\pi = I$.

Isomorphism theorems.

THM 1). if $\Phi: R \to S$ is a homomorphism of rings, then

$\ker\Phi$ is a two-sided ideal of R.

(more generally, Φ^{-1}(two-sided ideals of S) is a two-sided ideal of R)

$\Phi(R)$ is a subring of S and $R/\ker\Phi \cong \Phi(R)$ $\quad r+\ker\Phi \mapsto \Phi(r)$.

2) 在后.

3). Let $I \subset J$ be two-sided ideals of R,

then $J/I \subset R/I$ is a two-sided ideal of R/I and $(R/I)/(J/I) \cong R/J$.

4). Let I be a two-sided ideal of R.

Then there is a natural bijection

$$\{\text{left/right/two-sided ideals of } R \text{ containing } I\} \longleftrightarrow \{\text{left/right/two-sided ideals of } R/I\}$$
$$J \longmapsto J/I$$
$$\pi^{-1}(\bar{J}) \longleftarrow \bar{J}$$

preserving inclusion, sums, intersections and quotients.

Notation Let R be a commutative ring and let $\{a_j\}_{j\in J} \subset R$ be a subset.

Define $(a_j)_{j\in J} := \{\sum_{j\in J} r_j a_j \mid r_j \in R, r_j = 0 \text{ for all but finitely many } j\}$ called the ideal generated by $\{a_j\}_{j\in J}$

it is the minimal ideal containing $\{a_j\}_{j\in J}$.

eg. 1). $R=\mathbb{Z}$ $\quad (4,6)=\{4x+6y \mid x,y\in\mathbb{Z}\}=2\mathbb{Z}=(2)$.

2). $\Phi: \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ $\quad \ker\Phi = n\mathbb{Z} = (n)$.

3). R commutative $a \in R$

$\Phi_a: R[x] \to R$
$f(x) \mapsto f(a)$

$\ker \phi_a = \{f(x) \in R[x] \mid f(a)=0\} = (x-a)$.

$\Rightarrow R[x]/(x-a) \simeq R$.

4). R commutative $R[G]$ group ring

增广映射 augmentation map $\Phi: R[G] \to R$

$\sum_{g\in G} a_g g \mapsto \sum_{g\in G} a_g$

与研究群论的另一路径——同调代数相关.

$\ker \Phi = (g-g)_{g,g\in G} = (g-1)_{g\in G}$ is called the augmentation ideal of $R[G]$.

5). $R = R_1 \times R_2$ direct product of rings. $1_R = (1_{R_1}, 1_{R_2})$.

Then $R_1 \times \{0\}$ and $\{0\} \times R_2$ are ideals but usually not a subring.

Operations on rings.

Let I, J be two-sided ideals of R.

1). sum. $I+J := \{a+b \mid a\in I, b\in J\}$ two-sided ideal of R.

2). intersection. $I \cap J$ two-sided ideal of R.

3). product. $IJ := \{$finite sums of elements ab for $a\in I, b\in J\}$ two-sided ideal of R.

! NOT all elements of IJ are of the form ab.

eg. R commutative $I=(a_1,\cdots,a_s)$ $J=(b_1,\cdots,b_t)$.

then $I+J=(a_1,\cdots,a_s,b_1,\cdots,b_t)$. $IJ=(a_i b_j)_{\substack{i=1,\cdots,s\\ j=1,\cdots,t}}$.

$IJ \subset I\cap J \subset \begin{matrix} I \\ J \end{matrix} \subset I+J$

Second isomorphism theorem

Let S be a subring of R, I be a two-sided ideal of R

then $S+I := \{a+b \mid a\in S, b\in I\}$ is a subring of R, $S\cap I$ is a two-sided ideal of S

and $(S+I)/I \simeq S/(S\cap I)$.

REM. Meaning of quotient rings: imposing "relations" among generators.

eg. F field. $F[x,y,z]/(x-y^2, y-z^3) \simeq F[z]$. (强行让 $x=y^2$, 让 $y=z^3$.)

$x^2y = (x-y^2+y^2)^2 y = (x-y^2)\cdot * + y^5 = (x-y^2)\cdot * + (y-z^3+z^3)^5 = (x-y^2)\cdot * + (y-z^3)\cdot * + z^{15}$

So in the quotient ring $x=y^2$, $y=z^3$.